



Unit - I

Data - Data is a collection of raw, unprocessed facts, figures, symbols or observation that represent details about events, objects or phenomena.

- Data is the raw input i.e., stored in a database.

Types of Data in DBMS

1.) Structured Data

- Organized in tables (rows & columns)
- Ex:- SQL databases (MySQL, Oracle)

2.) Unstructured Data

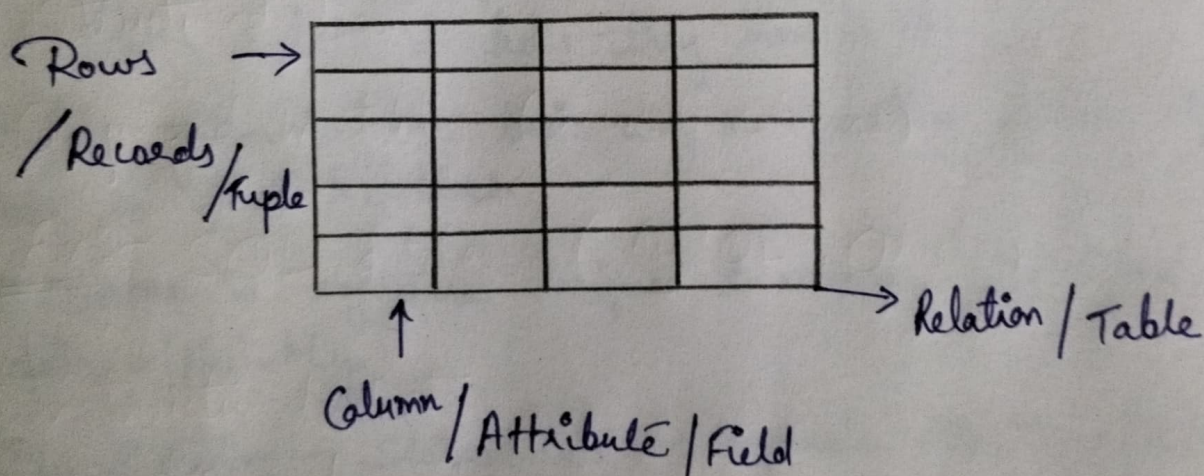
- No fixed format
- Ex:- Images, videos, emails

Summary :

3.) Semi-structured Data

- Partially Organized
- Ex:- JSON, XML files

- ★ Tables is also known as Relation.
- ★ Rows is also known as Records / Tuple.
- ★ Columns is also known as Field / Attributes.



Information — Data i.e., is processed to have meaning

- Meaningful data.

Summary :



DBMS - DBMS stands for database management system.

- A DBMS is a s/w system designed to manage & organize databases. It allows users to efficiently store, retrieve, update & manage data in an organized manner.

DBMS vs File System - A file system & a dbms are both used to store & manage data, but they differ a lot in how they handle it.

- A file system is the basic way an operating system stores data in files & folders.
- A DBMS is software designed to manage structured data efficiently.
- File System - stores data as separate files in folders.
- DBMS - Stores data in structured tables with rows & columns.

Summary :



S.No.	Features	File System	DBMS
1.	Data Organization	Files & folders	Tables / Structured
2.	Redundancy	High	Low
3.	Data Consistency	Poor	High
4.	Security	Basic	Advanced
5.	Query Support	None	SQL Supported
6.	Multi-user access	Limited	Strong
7.	Backup & recovery	Manual	Automatic
8.	Relationships	Not-Supported	Supported.

Advantages of DBMS

Due to its centralised nature, the database system can overcome the disadvantages of the file system-based system

- Concurrent Use — A database system provides facility to access database several users concurrently. Let we understand it by taking an example that a movie online booking system database employee of differently branches concurrently access the database. Each employee

Summary :

can handle customers at their desk, able to see the seats available for the booking from the interface provided to them.

- Structured data & details - One of the essential features of the database system is not only to store the data & to provide access to the database. Each employee can handle customers at their desk. However, it also provides details about the data & how to access & use it. Taking an example when you are going for an online form of exam then details about every field is given, what types of data it accepts & format details also.
- Data Independence - Another essential characteristics of the DBMS, is data independence in which application through which user can interact with the user is not dependent on the physical data storage. So, if there is any change in the appⁿ program, it will not affect the data stored in the database.

- The integrity of data — This characteristic of the DBMS deals with one of the fundamental properties of data called integrity, in which if some data is saved in the database & later retrieved from the database it must be same. This also covers restriction on the unauthorised access of the data that can make changes in the data sets.
- Transaction of Data — A transaction is a set of actions that are done in a database to transfer it from one consistent state to another. If it is not handled correctly, it may result in inconsistent state & loss of data, so DBMS has certain constraints to maintain the fundamental like atomicity, integrity, isolation & durability.
- Data Persistence — This is one of the essential characteristics of the database because data can be retained in the database for years & it should be in the same condition. In the banking system, a user will open his account & lifelong maintain it, or a user has opted an LIC policy, or media claim policy.

Summary :

DBMS Architecture

DBMS architecture dictates how users & appⁿ interact with a database. It is broadly categorized into two models : System Structure layers (defining internal processing components) & user-tier models (defining the client-server relationship).

- 1.) The 3-level ANSI-SPARC Architecture — This internal framework organizes the DBMS into distinct layers to achieve data independence, meaning changes at one level do not require alterations at higher levels.
- External Level (View level) : The highest level. It describes only the part of the database that a specific user group cares about, hiding the rest of the database for security & simplicity.
 - Conceptual level (Logical level) : The middle level. It describes what data is stored in the database & the relationships among the data. It serves

Summary :

as the global view for the entire organization.

- Internal Level (Physical level) — The lowest-level. It details exactly how data is physically stored on the hardware (e.g., storage allocation, indexing, & compression).

Q.2) DBMS Processing Components — Beneath the layers, the core DBMS engine coordinates data access using several internal modules:

- Query Processor :- Parses, compiles & optimizes user queries (SQL) into a low-level execution plan.
- Storage Manager :- Interfaces with the underlying operating system & physical storages. It includes:
 - (a) Authorization & Integrity Manager : checks if users have permission to access the data.

Summary :



(b) Transaction Manager : Ensures ACID properties (Atomicity, Consistency, Isolation, Durability) by handling rollbacks and concurrency.

(c) Buffer Manager : Caches frequently accessed data in main memory to speed up performance.

(d) File Manager : Allocates space on the physical disk.

3) Client - Server Architectures — These structures determine where the application logic & user interfaces physically reside in relation to the database.

- 1-Tier Architecture — The client, application & database all reside on the same machine (e.g., MS Access). Best for standalone or personal use.

Summary :

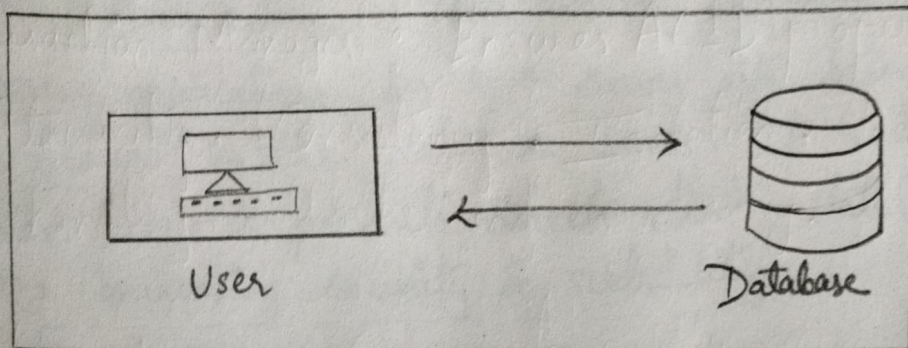


Fig:- 1-Tier Architecture

- 2-Tier Architecture — A basic client-server model.

The client handles the user interface & appⁿ programs, directly communicating with the database server via APIs like ODBC/JDBC.

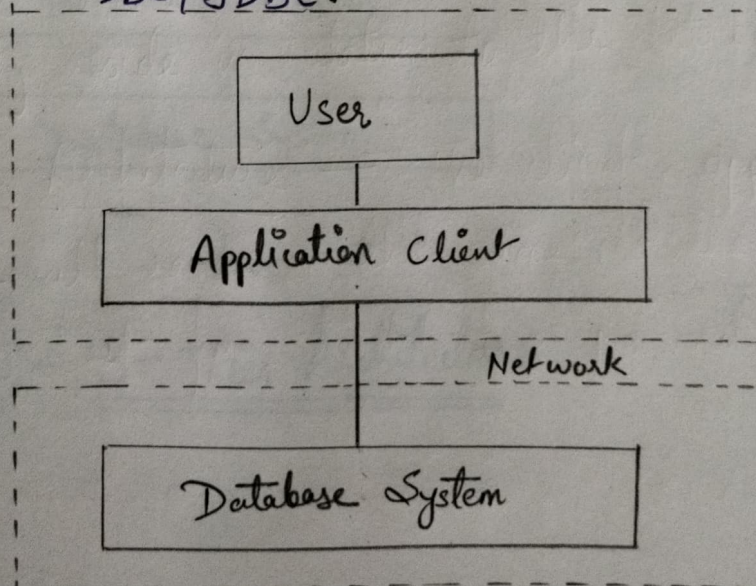


Fig:- 2-Tier Architecture

Summary :

- 3-Tier Architecture — The industry standard for web & enterprise applications. An Appⁿ Server sits between the user (client) & the database server. It abstracts the business logic & prevents direct client - database interaction, increasing security & scalability.

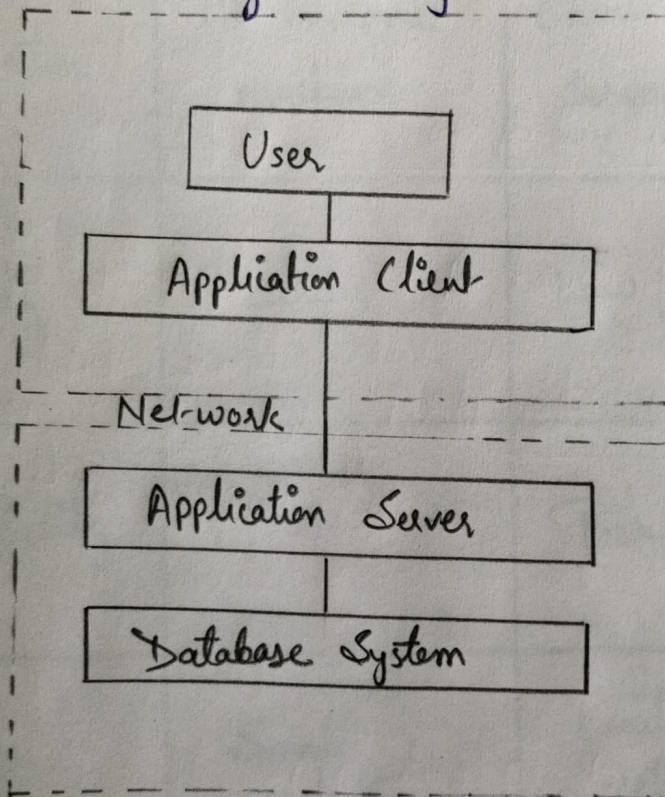


Fig:- 3-Tier Architecture

Summary :



Key Difference -

	Features	1-Tier	2-Tier	3-Tier
1.)	Structure	All on one machine	Client + Server	Client + App Server + Data Server
2.)	Security	Low	Low (Direct access)	High (Indirect Access)
3.)	Scalability	Poor	Poor	Excellent
4.)	Usage	Small projects	Local Networks	Web / Large App ^s .

Summary :



Some One Word Answers:-

Q.1.) In which architecture is the entire database system located on a single machine?

A.1.) One-Tier Architecture

Q.2.) In Two-Tier Architecture, the client communicates directly with?

A.2.) Database Server

Q.3.) Which architecture provides better security & scalability?

A.3.) Three-Tier

Q.4.) Which layer in 3-tier architecture communicates with the database directly?

A.4.) Data Layer

Q.5.) Three-Tier Architecture consists of:

A.5.) Client, Appⁿ Server, Data Server.

Summary :

Data Models

- A data model in dbms is a collection of concepts, tools & rules used to describe the structure of a database.
- It defines how data is stored, organized, related & manipulated.
- In simple terms, a data model is like a blueprint for designing a database.
- A database model is a formal framework that defines how data is structured, related, & constrained inside a database system. It specifies how entities, attributes & relationships are organized at the logical level. Database models provide structural rules for designing schemas before physical implementation.

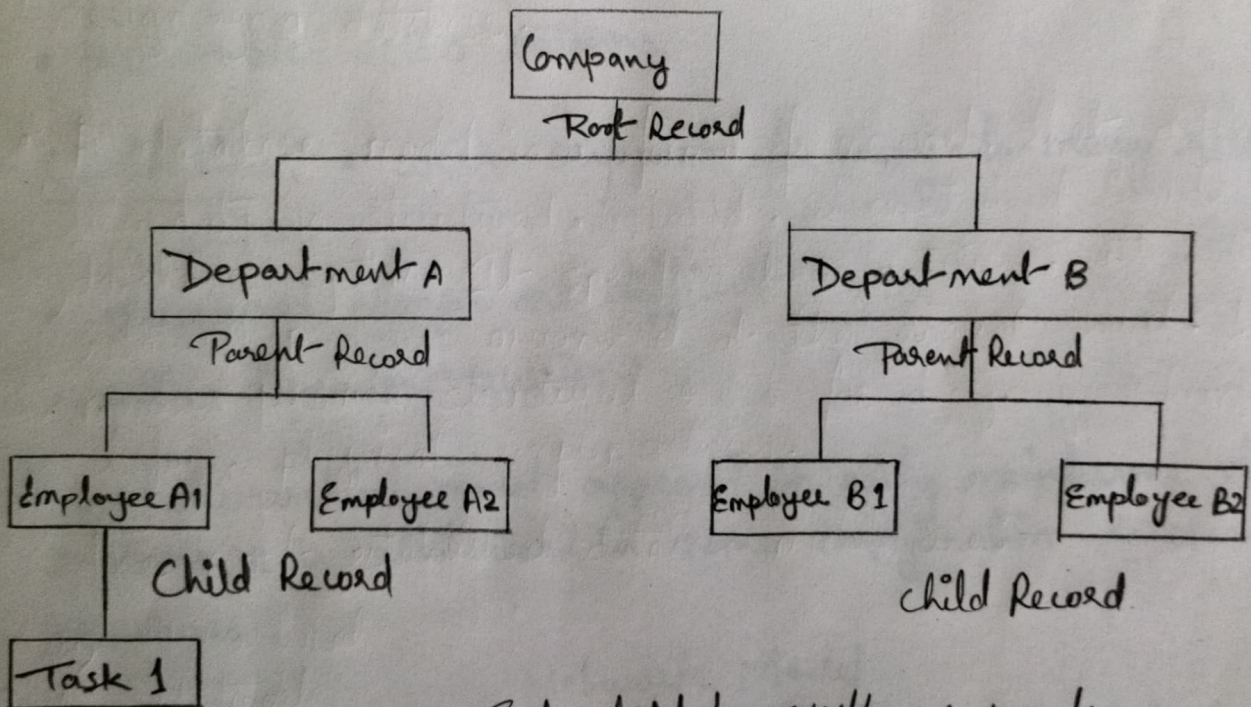
The most established database models are:—

- a) Hierarchical
- b) Network
- c) Relational
- d) Entity - Relationship
- e) Object - oriented
- f) Object - relational.

Summary :

Hierarchical Model

- The hierarchical model organizes data in a tree structure. Each record has a single parent & can have multiple child record. Relationships follow a strict one-to-many pattern.
- Data is stored as nested parent-child segments. Access paths are predefined & traversal follows the tree from root to leaves.



Each child has exactly one parent.

Summary :

Structure Characteristics :

- Tree-shaped hierarchy

- single parent per child
- fixed navigation paths
- strong structural rigidity

Strengths :

- Simple structure

- fast traversal along known paths
- predictable access patterns

Limitations :

- weak support for many-to-many relationships.

- structural changes are difficult.
- path-dependent queries.

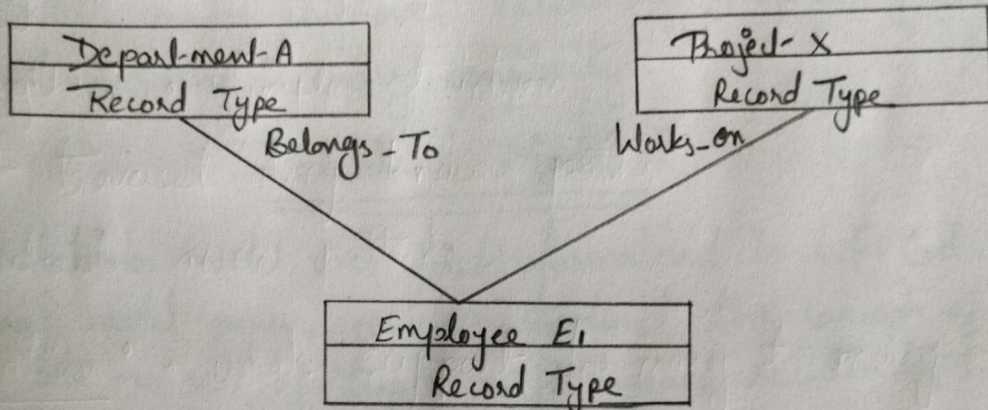
The hierarchical model appeared in early mainframe database systems & is still used in some configuration & directory structures.

Network Model

The network model extends the hierarchical model by allowing records to have multiple parents. It represents data as a graph structure rather than a strict tree.

Summary :

Relationships are defined through explicit link sets between record types. A record can participate in multiple relationships simultaneously.



A record can participate in multiple relationship sets (Multiple "parents").

Structure Characteristics :

- graph-based structure

- multiple parent relationships
- pointer-based navigation
- predefined relationship sets

Strengths :

- supports many-to-many relationships

- more flexible than hierarchical model
- efficient for complex relationship graphs.

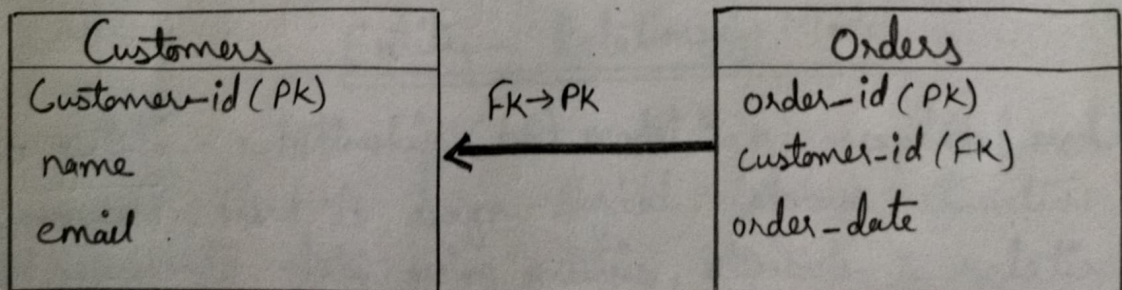
Summary :

- Limitations:
- complex schema design
 - navigation logic is tightly coupled to structure
 - difficult query formulation.

The network model improved relationship flexibility but increased modeling & query complexity.

Relational Model

- The relational model was first formalized in E.F. Codd's relational model paper, which introduced the concept of representing data as sets of tuples with defined relationships.
- The relational model represents data as tables composed of rows & columns. Relationships are defined through keys & constraints rather than physical pointers.
- Each table represents an entity or relationship. Rows represent records. Columns represent attributes. Relationships are enforced through primary & foreign keys.



Summary: Here, PK → Primary key
FK → Foreign key

Structure characteristics :

- table-based structure

- key-driven relationships
- declarative query model
- set-based operations

Strengths :

- strong mathematical foundation

- flexible query capabilities
- clear constraint enforcement
- model independence from access paths.

Limitations :

- joins can be expensive at scale

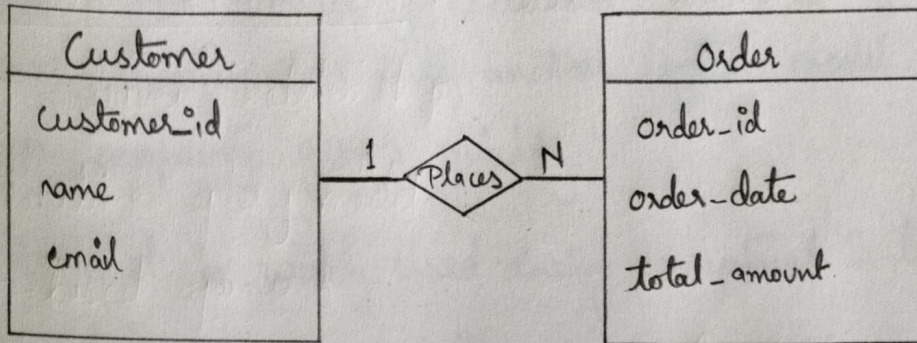
- complex object structures require decomposition
- strict schema requirements

The relational model remains the dominant logical model in database design & underpins SQL-based systems.

Entity - Relationship Model

The entity-relationship (ER) model is a conceptual modeling framework used to design logical database structures. It represents data using entities, attributes & relationships.

Summary :



Cardinality: One Customer can place many Order (1:N).

It is primarily a design model, not a storage model. ER diagrams are used to define structure before converting it into a relational or other logical schema.

Structural Characteristics:

- entity types

- attribute sets
- relationship types
- cardinality rules

Strengths:

- clear visual representation

- strong design ~~communication~~ communication tool
- supports early modeling stages
- maps well to relational schemas.

Summary :

- Limitations :
- not directly executable
 - requires transformation into another logical model
 - diagram complexity grows quickly

— The ER model is widely used during conceptual & logical database design.

Object-Oriented Model

The object-oriented model stores data as objects that combine attributes & behaviour. Objects are grouped into classes & support inheritance & encapsulation.

Relationships are represented through object reference rather than foreign keys.

Structure Characteristics :

- Object identity
- Class hierarchies
- inheritance
- method binding

Strengths :

- natural fit for object-oriented applications.
- Supports complex data structures

Summary :



- reduces object - relational mapping layers.

Limitations :

- weaker standardization

- complex query models
- limited adoption compared to relational systems

This model is used in object database systems & specialized applications.

Object - Relational Model

The object - relational model extends the relational model with object - oriented features. It keeps table structures while adding support for complex types & inheritance.

It combines relational foundations with richer data representations.

Structure Characteristics :

- relational tables

- user-defined types
- nested structures
- inheritance support

Strengths :

- compatible with relational querying

- supports complex data types

Summary :

- reduces schema fragmentation

Limitations :

- increased schema complexity

- vendor-specific extensions
- portability differences

Object-relational modeling appears in advanced relational systems that support extended type systems.

Schemas & Instances

Schema - A schema is the structure or design of the database.

- It is like a blueprint of the database.

Instance - An instance is the actual data stored in the database at a particular time.

- It changes whenever data is added, deleted or updated.

Schema	Instance
• Database design/structure • Changes rarely • Blueprint	• Actual data • Changes frequently • Current content.

Summary : Note :- Schema are 3 types :-

- 1) Physical Schema - The design of a database at the physical level is c/d physical schema.
 - 2) Logical Schema - Design of database at a logical level is c/d logical schema.
 - 3) View Schema - Design of database at a view level is c/d view schema.
- Here c/d is called.



Data Independence

The ability to modify schema definition in one level without affecting schema of that definition in the next higher level is called data independence. ~~There are~~ There are two levels of data independence, they are Physical data independence & Logical data independence.

Physical data independence :- Physical data independence is the ability to change how a database stores its data on physical hardware - such as altering file structures, storage devices, or indexing methods - without requiring any modifications to the database's logical design or the appⁿ programs that access it.

Logical data independence :- Logical data independence is the ability to change the overall logical structure of a database (conceptual schema) without altering user interfaces, external views, or requiring existing appⁿ programs to be ~~new~~ rewritten.

It protects the appⁿ layer from structural database updates, ensuring long-term flexibility & easier system maintenance.

Summary : Note :- Logical data independence is harder to achieve than physical data independence.

Functions of DBA & Designer

- A Database Administrator (DBA) is responsible for managing, maintaining, securing & optimizing an organization's databases.
- Their primary goal is to ensure that data is safe, highly available, & easily accessible to authorized users & appⁿ at all times.

Key Functions of a DBA.

1.) Database Installation & Configuration

- Install, configure & maintain the database management system.
- Setup database software, storage structure & user environments.

2.) Database Design

- Collaborate with designers to create logical & physical database designs.
- Ensure the database schema meets organizational requirements.

Summary :

3.) Backup & Recovery

- Develop & implement backup policies & disaster recovery strategies.
- Regularly perform backups & test recovery procedures to prevent data loss.

4.) Performance Optimization

- Monitor & improve database performance using indexing, query optimization, & resource management.
- Analyze query execution plans & suggest improvements.

5.) Security Management

- Define & enforce database access policies to ensure data confidentiality & integrity.
- Manage user roles, permissions, & authentication mechanisms.

6.) Capacity Planning

- Plan & manage database storage & growth to ensure scalability.
- Monitor disk usage, memory allocation & hardware resources.

Summary :



7.) Database Maintenance

- Regularly update & patch the database software.
- Manage data integrity by running consistency checks & resolving anomalies.

8.) User Support & Training

- Provide support to developers, users, & other stakeholders in database operations.
- Offer training on database tools & query languages.

9.) Troubleshooting

- Diagnose & resolve database issues such as performance bottlenecks, crashes or corruption.
- Monitor logs & system alerts for potential problems.

10.) Documentation

- Maintain comprehensive documentation of database structures, configuration & processes.
- Ensure proper record-keeping for audit & compliance purposes.

Summary :

- The Database Designer focuses on planning & structuring the database to meet organizational requirements. Their primary role is to create an efficient, logical & scalable design for storing & managing data.

Key functions of a Database Designer

1.) Requirements Analysis

- Interact with stakeholders to understand the data requirements, workflows & business rules.
- Document data use cases, relationships & constraints.

2.) Conceptual Design

- Develop a high-level conceptual schema using tools like Entity-Relationship (ER) diagrams.
- Define entities, attributes & relationships without considering the physical implementation.

3.) Logical Design

- Convert the conceptual schema into a detailed logical schema.
- Normalize tables to eliminate redundancy & ensure data integrity.
- Define relationships, keys & ~~on~~ constraints.

Summary :



4.) Physical Design

- Design the physical schema that specifies how the data will be stored in the database.
- Plan storage structures, indexing strategies & partitioning for optimal performance.

5.) Data Modeling

- Create data models that represent the database structure at different levels (Conceptual, logical & physical).
- Use data modeling tools to visualize & refine the database design.

6.) Collaboration with DBA.

- Work closely with the DBA to implement & test the database design.
- Provide feedback to align the design with performance & scalability requirements.

7.) Optimization

- Optimize the schema for performance, considering factors like query frequency & data size.

Summary :



- Suggest improvements to indexing, partitioning or denormalization when necessary.

8.) Documentation

- Document the design, including diagrams, schemas, & design decisions.
- Ensure the design is well-documented for implementation & future maintenance.

9.) Testing

- Validate the database design by running test queries & scenarios.
- Ensure the design meets functional & non-functional requirements like reliability & scalability.

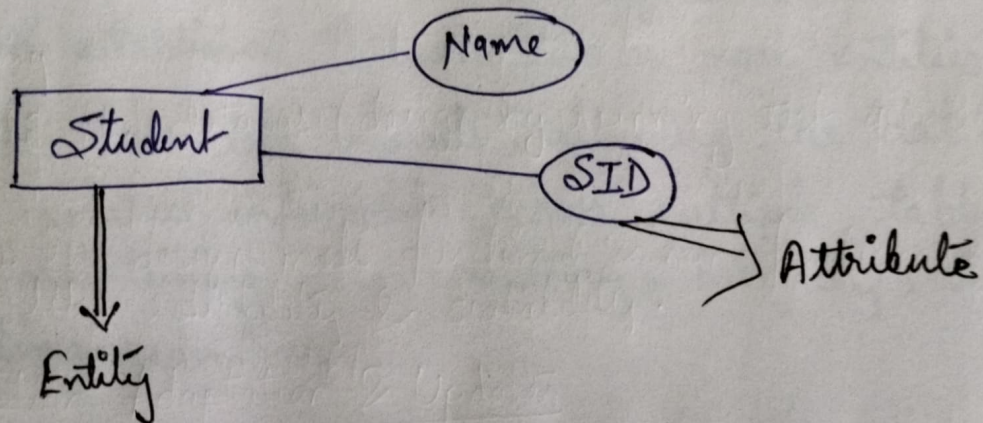
10.) Adoption Adaptation & Updates

- Revise the design based on feedback or changing business requirements.
- Ensure the database design evolves with the organizations needs.

Summary :

Entities & attributes

- ⇒ An entity is a real-world object, concept or event (like a person, place, car or bank account) about which data is stored.
- ⇒ Attributes are the characteristics, properties, or traits that define & describe an entity.



Entity types

- ⇒ In a Database Management System (DBMS), an entity is a real-world object, concept or piece of data that ~~be~~ can be uniquely identified & stored.

There are two primary categories of entities; strong entity & weak entity both of these are essential to structure the database.

Summary :

Key attributes

In a DBMS, a key attribute is a property or column whose values are unique for every entity (row) in a table, allowing for individual records to be identified distinctly.

Relationships

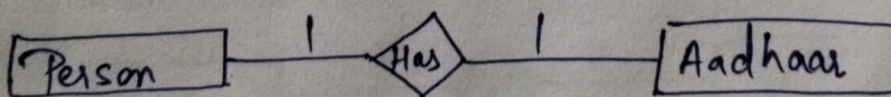
A relationship in a DBMS is a logical association or connection established between two or more entities (tables).

It dictates how data interacts, allowing the database to link related information across multiple tables while maintaining data consistency, integrity, & preventing redundancy.

The 4 Primary Types of Relationships

1) One-to-one (1:1) :- One record in Table A is associated with exactly one record in Table B.

It is also written as 1:1.

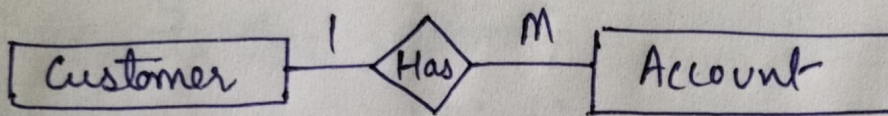


Summary :

Note: Relationship has 2 things being
by entity

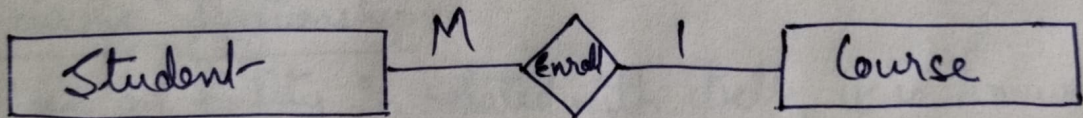
2.) One-to-many (1:M) :- One record in Table A is associated with multiple records in Table B, but records in Table B only link to a single record in Table A.

It is also written as 1:M.



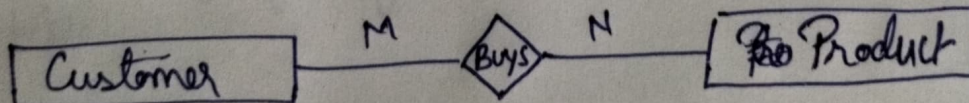
3.) Many-to-one (M:1) :- Multiple records in Table A map to a single record in Table B.

It is also written as M:1.



4.) Many-to-many (M:N) :- Multiple records in Table A can be associated with multiple records in Table B.

It is also written as M:N.



Summary :



A relationship describes relations between entities. The relationship is represented using diamonds.

There are 3 types of relationship that exist between Entities.

1.) Binary Relationship

- a.) One to One relationship
- b.) One to Many relationship
- c.) Many to One relationship
- d.) Many to many relationship

2.) Recursive Relationship

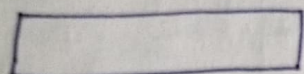
When an Entity is related with itself, it is known as Recursive Relationship.

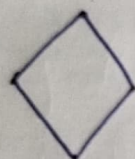
3.) Ternary Relationship

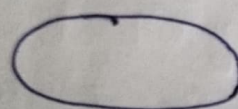
Relationship of degree three is called Ternary relationship.

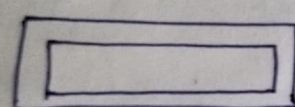
Summary :

E-R Diagram — This is also known as entity relationship diagram.

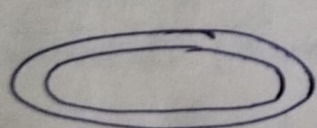
 $\Rightarrow$ Entity

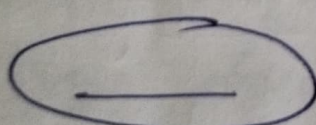
 $\Rightarrow$ Relationship

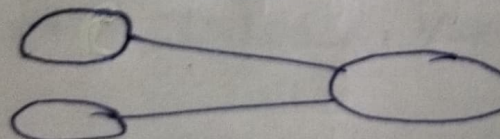
 $\Rightarrow$ Attribute

 $\Rightarrow$ Weak Entity

 $\Rightarrow$ Weak Entity relationship

 $\Rightarrow$ Multivalued Attribute

 $\Rightarrow$ Key Attribute

 $\Rightarrow$ Composite Attribute

Summary :

Entity— Entities are represented using rectangle. Rectangles are named with the entity set they represent.

Ex:- Student, Faculty, Course.

- Strong Entity :— This type of entities has key attributes set that are used to create a primary key. Like student entity has Roll No as a critical attribute.
- Weak Entity :— This type of entities does not have any primary key attribute & they are dependent on some other entity has the crucial prime attribute. For example, Employee child is a weak entity some other entity has the crucial prime attribute.
- Composite Entity :— This type of entities is used to replace many to many relationships, & Entity replaces that relationship. It is of three types one-to-one cardinality, one-to-many cardinality, many-to-many cardinality.

Attribute— Attribute are the properties of entities. Attribute are represented using ellipses. Every ellipse represents one attribute & is directly connected to its entity (rectangle).

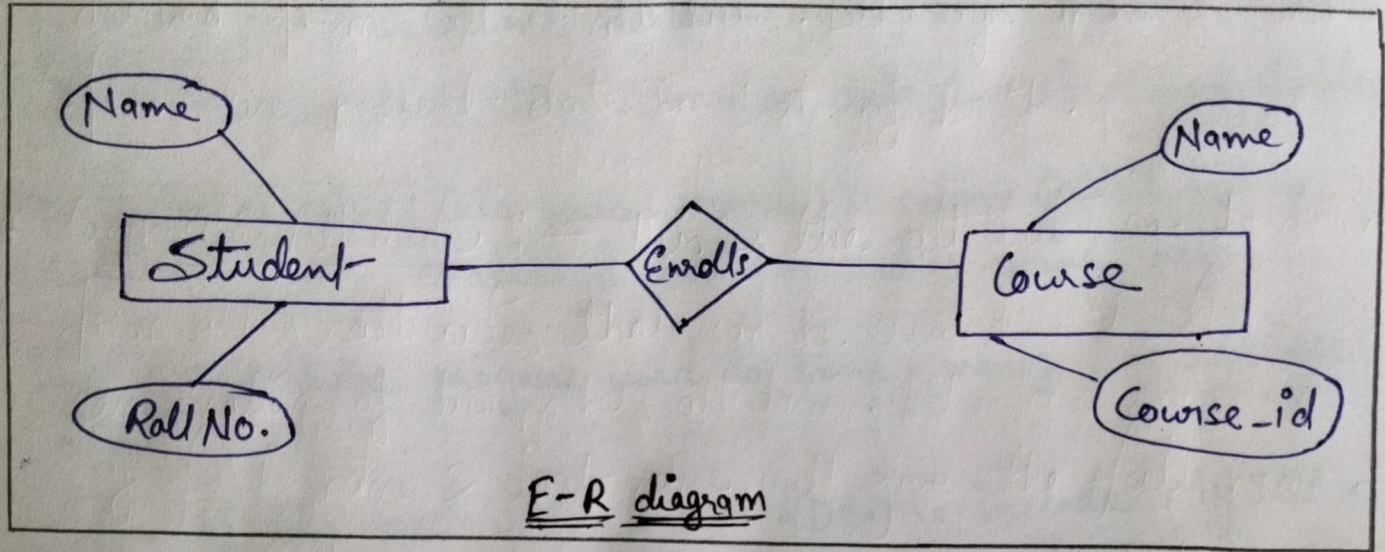
Summary :



- Derived attributes :— This is the attributes that can be derived from other attributes. Like age can be derived from the present date & Date of Birth.
- Composite Attributes :— This is the attributes can be derived from other attributes that can be created with the combination of two other attributes. Like a combination of first name & last name will give the full name.
- Single-value attribute :— This is the attributes which have single value when they are converted in the form of a table. Like DOB of a person.
- Multi-valued attribute :— This is the attributes which have multiple values for the instance when an entity is converted to a table, & that attribute allows multiple values for it. Like a person have two contact numbers.

Summary :

Defining the E-R diagram of database.



Here, $\bigcirc \Rightarrow$ Attribute

$\square \Rightarrow$ Entity

$\diamond \Rightarrow$ Relationship

This ER diagram shows the entities, attributes & relationships in a database. Here a student enrolls in a course & both entities contain their own attributes.

$\Rightarrow$ An E-R diagram is a visual model that represents entities, their attributes & the relationships between them in a database.

Summary :